Multimodal Emotion Recognition under Realistic Evaluation: Insights from IEMOCAP

A Protocol-Aware Comparative Study

Project Report

April 2026

Dataset	Utterances	Speakers	Best Macro F1
IEMOCAP	**7,433**	**10**	**82.62%**

Main research question: Does multimodal emotion recognition improve speaker-independent generalization, or do strong acoustic baselines remain dominant under realistic evaluation?

Abstract

This report investigates multimodal speech emotion recognition on IEMOCAP under realistic speaker-aware evaluation. The study is motivated by a previous RAVDESS analysis, where random split performance was shown to be optimistic because of speaker leakage. Audio-only, text-only, and multimodal systems are compared under GroupKFold and Leave-One-Speaker-Out (LOSO) protocols. The main result is that multimodal fusion remains competitive but does not provide a stable advantage over strong audio baselines under strict speaker holdout. The analysis therefore emphasizes protocol sensitivity, effect sizes, and class-level error structure rather than isolated best scores.

1. Context and Research Motivation

Speech emotion recognition is challenging because emotional states are expressed through both acoustic and contextual cues, while speaker identity introduces substantial variability. High reported performance can therefore be misleading when evaluation protocols do not sufficiently control for speaker overlap.

This concern became explicit in the earlier RAVDESS project, where performance decreased substantially under speaker-independent validation. The present IEMOCAP study extends that investigation to a more realistic conversational setting and tests whether multimodal modeling can mitigate this limitation.

2. Objective of the Study

The objective of this report is to assess whether multimodal fusion provides meaningful and stable gains over audio-only baselines on IEMOCAP when evaluation is performed under speaker-aware constraints.

More specifically, the study aims to:

- quantify the effect of evaluation protocol on reported performance,
- compare audio-only, text-only, and multimodal systems,
- analyze whether fusion gains remain valid under strict speaker holdout,
- and interpret the dominant residual error patterns.

> The main scientific criterion retained throughout the report is the distinction between optimistic random-split performance and speaker-independent generalization.

3. Dataset Description

Experiments are conducted on the KaggleHub release of IEMOCAP. Each utterance is associated with aligned acoustic features, transcript text, emotion label, speaker identifier, and dialogue metadata.

Compared with RAVDESS, IEMOCAP is more challenging because it contains broader conversational variability, both scripted and improvised interactions, and less exaggerated emotional expression.

3.1 Dataset Summary

Property	Value
Total utterances	7,433
Number of speakers	10
Emotion classes	6 (angry, excited, frustrated, happy, neutral, sad)
Dialogues	151
Dialogue types	3,967 improvised; 3,466 scripted
Gender distribution	3,939 male; 3,494 female
Audit result	No issues detected; no samples removed

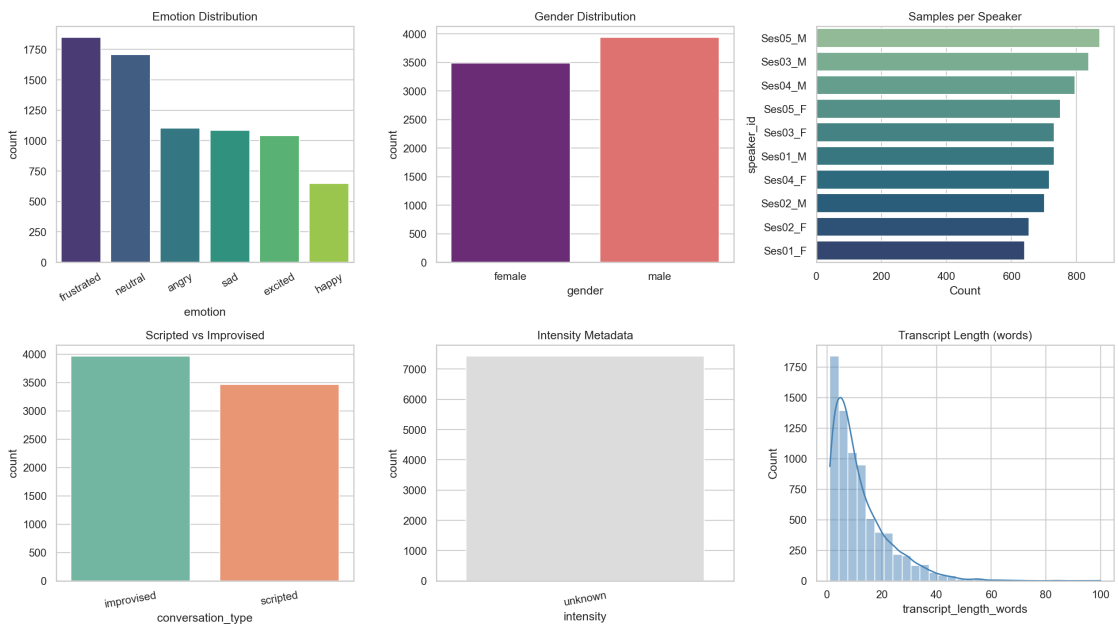

Figure 1: Exploratory overview of IEMOCAP: class distribution, gender distribution, speaker distribution, dialogue type, and transcript-length statistics.

The dataset is appropriate for this study because it allows evaluation of whether models generalize emotion structure across speakers rather than simply benefiting from repeated exposure to the same voices.

4. Methodology and Pipeline

4.1 Modalities and Representations

Three modality settings are evaluated: audio-only, text-only, and multimodal (audio + text).

Audio is represented using concatenated precomputed acoustic blocks (`audio_features` + `auxiliary_features`). Text is represented through TF-IDF, dense semantic embeddings, and stronger transformer-based embeddings in the extended analysis.

4.2 Fusion Strategies

Three fusion families are compared:

- **Early fusion**: concatenation of audio and text features in a shared vector space.
- **Adaptive weighted late fusion**: combination of modality-level probabilities with learned reliability weighting.
- **Cross-modal interaction fusion**: interaction-aware joint representation using multiplicative and distance-based terms.

4.3 Models

The study evaluates Random Forest, MLP, and a lightweight 1D CNN baseline over audio vectors. The CNN is included as a reference deep model rather than as a fully optimized end-to-end architecture.

5. Experimental Setup

5.1 Evaluation Protocols

Two speaker-aware protocols are retained as primary references:

- **GroupKFold (k=5)** for stable grouped estimation,
- **Leave-One-Speaker-Out (LOSO)** for strict speaker-level transfer.

5.2 Metrics

The report uses Accuracy, Balanced Accuracy, and Macro F1, with emphasis on Macro F1 to better reflect class-balanced performance.

5.3 Hyperparameter Selection

Hyperparameters are optimized with group-aware cross-validation in order to avoid leakage during model selection.

6. Results and Interpretation

6.1 Protocol Sensitivity

A major finding is the size of the gap between optimistic and speaker-aware evaluation.

Configuration	Standard Split	Speaker-Independent Split	Difference
Audio-only (MLP)	0.8196	0.6989	-0.1208
Audio + dense text (MLP)	0.8294	0.7031	-0.1263

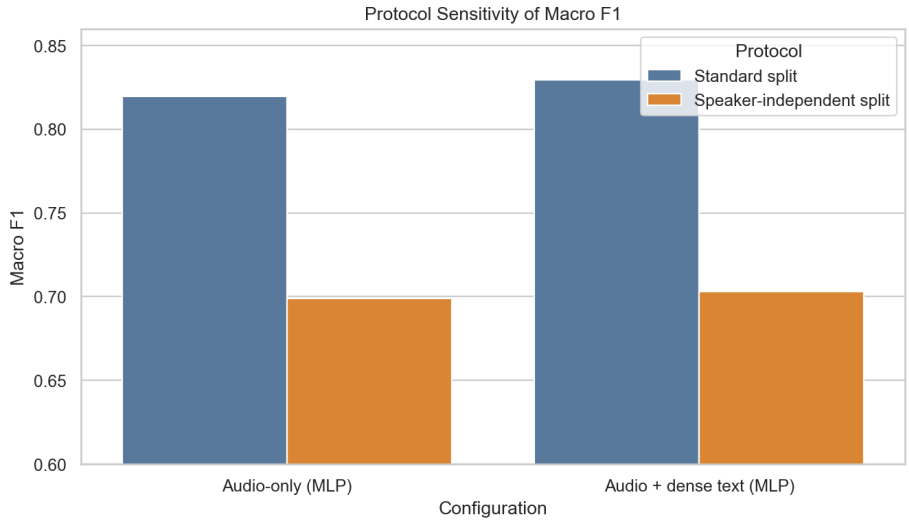

Figure 2: Macro F1 under standard and speaker-independent evaluation. The protocol gap quantifies the extent of evaluation inflation under random splitting.

These values show that random splits materially inflate expected performance. Both audio-only and multimodal settings benefit from speaker-correlated information under easier splitting.

6.2 GroupKFold Baselines

The strongest GroupKFold settings are reported below.

Rank	Configuration	Macro F1
1	Audio only + MLP	0.8257
2	Audio + dense text + RandomForest	0.8227
3	Audio + TF-IDF + RandomForest	0.8162

Text-only baselines remain much lower (0.4643 for TF-IDF + RandomForest and 0.3901 for dense text + RandomForest), indicating that acoustic information remains the dominant discriminative channel in this pipeline.

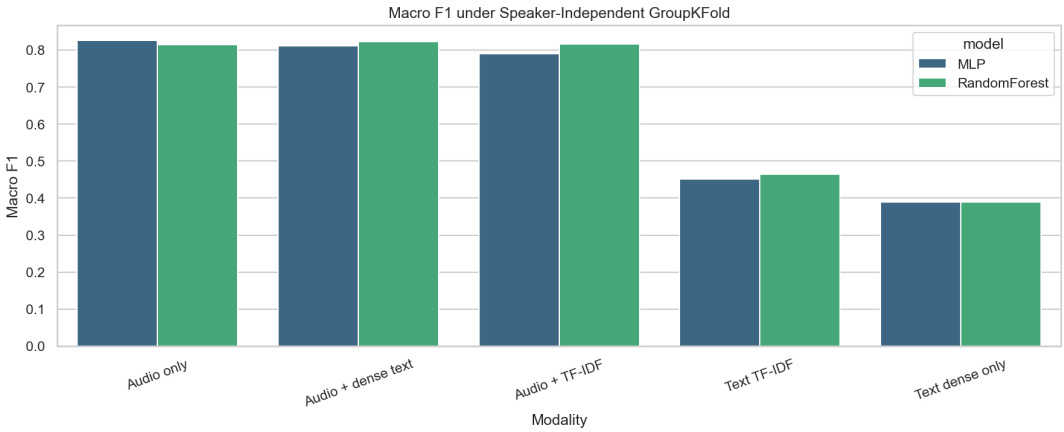

Figure 3: GroupKFold Macro F1 comparison across modality-model pairs.

6.3 Extended Fusion and LOSO

Extended evaluation confirms that multimodal gains are narrow and protocol-dependent.

- Best GroupKFold: Adaptive weighted late fusion + MLP = 0.8262
- Best LOSO: Audio only (LOSO) + MLP = 0.8222
- Best multimodal LOSO: Adaptive weighted late fusion (LOSO) + MLP = 0.8165

The best multimodal gain over the best audio baseline in GroupKFold is only +0.0005, while in LOSO the best multimodal result falls below the best audio baseline by -0.0056.

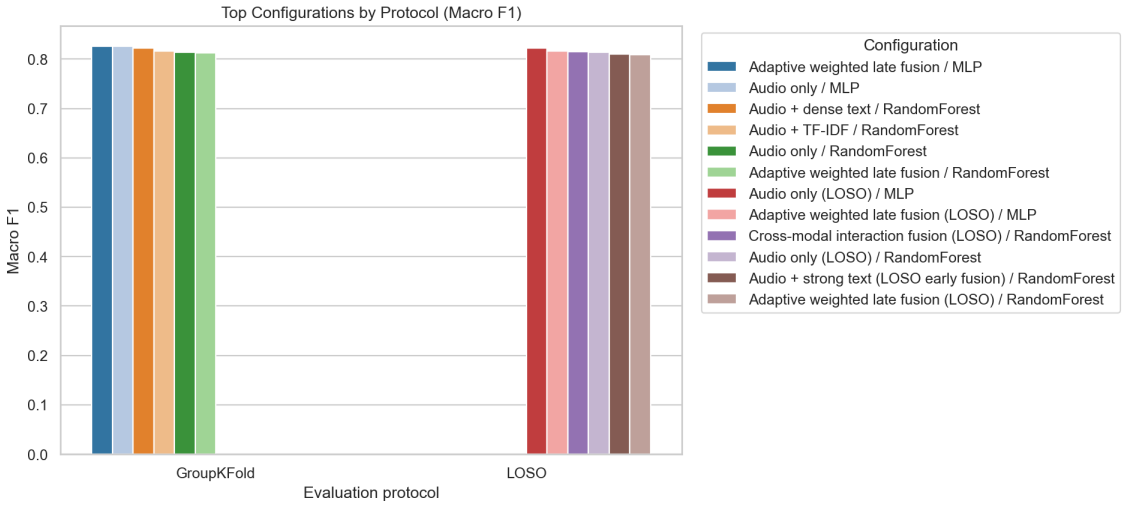

Figure 4: Top-performing configurations under GroupKFold and LOSO. Multimodal gains remain small and are not preserved uniformly across protocols.

6.4 Error Structure

The confusion profile is dominated by adjacent affective states such as neutral versus frustrated and happy versus excited. This indicates that future progress is more likely to come from better boundary-level disambiguation than from simple feature concatenation.

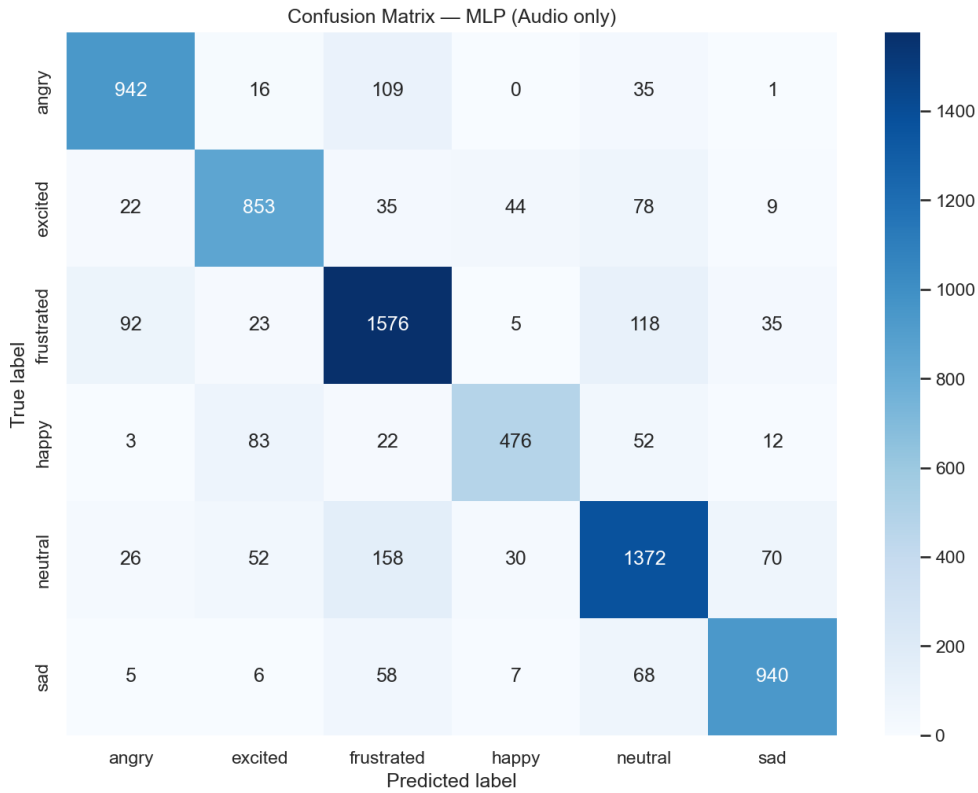

Figure 5: Confusion matrix of the selected best baseline. Adjacent affective classes remain the dominant source of error.

7. Link with the RAVDESS Project

The continuity between both projects is methodological. RAVDESS identified the risk of speaker leakage under random splits. IEMOCAP then tested whether multimodal information reduces this problem in a more realistic conversational setting.

The current results show that multimodality does not automatically resolve speaker-level generalization. IEMOCAP behaves differently from RAVDESS because conversational variability is broader, features are more compact and preprocessed, and emotional expressions are less exaggerated.

The transferable lesson across both projects is therefore not the exact performance value, but the need for protocol-aware and effect-size-based interpretation.

8. Positioning with Recent Research

Recent multimodal SER research points in the same general direction as this project:

- transformer-based cross-modal attention often improves interaction modeling compared with simple concatenation,
- self-supervised speech representations such as wav2vec 2.0 and HuBERT strengthen acoustic transfer,
- modality-reliability and uncertainty-aware fusion are increasingly preferred over fixed fusion rules,
- and robust evaluation protocols are now emphasized more strongly than random-split benchmark reporting.

This project aligns with that methodological shift. Its results also support a broader conclusion found in recent work: multimodal fusion is not inherently superior unless the representation and fusion mechanisms are well matched to the task.

9. Discussion

Several factors explain why multimodal gains remain limited in the current pipeline:

1. transcript content often encodes emotion only indirectly,
2. acoustic cues remain the dominant emotion carrier,
3. fusion depth is still moderate,
4. and generic semantic embeddings are not explicitly optimized for emotion discrimination.

The main contribution of the present study is therefore methodological rigor: reliable SER conclusions require speaker-aware evaluation, effect-size analysis, and structured error interpretation.

10. Conclusion and Future Work

Two conclusions are strongly supported by the evidence:

1. evaluation protocol substantially changes performance interpretation,
2. and multimodal fusion does not provide stable superiority over strong audio baselines under strict speaker-aware testing in this setup.

Future work should prioritize attention-based fusion, sequence-aware modeling, emotion-oriented fine-tuning of text encoders, per-speaker variance reporting, and cross-dataset transfer analysis.

References

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- Hsu, W.-N., et al. (2021). HuBERT: Self-Supervised Speech Representation Learning by Masked Prediction of Hidden Units.
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- Poria, S., Cambria, E., Hazarika, D., and Vij, P. (2017). A Deeper Look into Sarcastic and Emotion Recognition in Conversation.
- Recent IEMOCAP multimodal benchmark studies emphasizing speaker-aware evaluation.